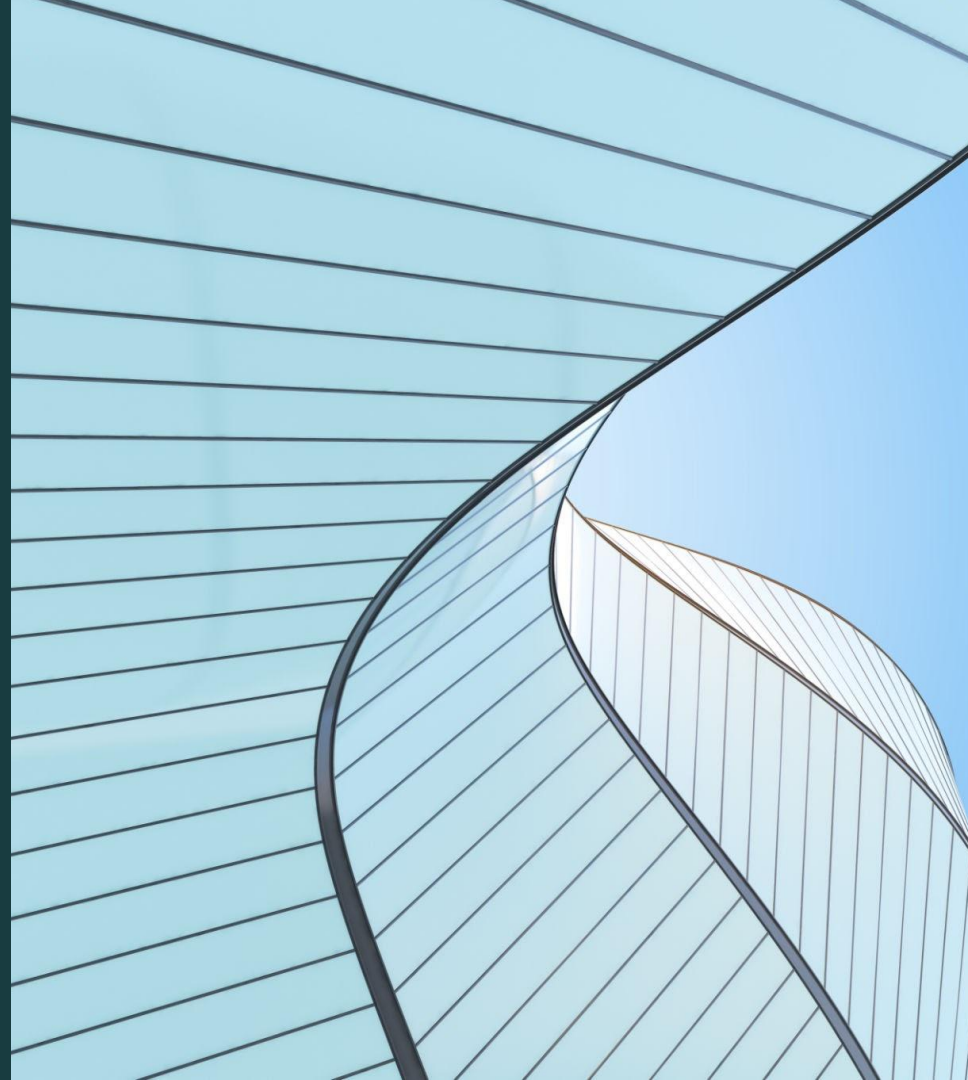


Ai-vian



By: Joanne He



Content Overview

1

Motivation

2

Dataset(s)

3

Potential Models



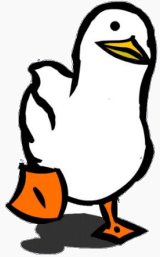
4

Predictions/Challenges

5

Conclusion/Summary

Motivations:



- ❖ Create a tool to identify birds on campus
- ❖ Enable curiosity about natural surroundings
- ❖ Enhance the experience of going on walks, bird watching, etc.



Datasets:



Xeno-Canto Avian Vocalizations (A-Z)

Dataproz

A lot of data is readily labeled and available

- 548 species labeled
- Range of 1 – 200+ recordings per
- Approximately 30gb
- Data is well labeled

Birdsongs from Europe (Xeno-Canto)

- 2150 files of various european species
- Data labels are consistent with other dataset
- Gender is clearly labeled and audio quality is consistent

Avian Vocalizations from CA & NV, USA

- Specifically pertains to birds from CA and NV in the US
- 2730 files of audio clips
- Data labels are compatible with aforementioned datasets.

Datacons

Lack of variation in data sources

- Gender isn't clearly labeled
- A lot of data for one person to go through

- Also from xeno-canto
- May have overlapping data with the A-Z dataset previously mentioned

- May have overlapping data with a-z dataset
- Gender is not clearly defined

Potential Models:

PANNs/YAMNets/VGGish – CNN (convolutional neural networks) Based Models

Calculates spectrograph data from audio recordings varies in purpose: General, AudioClassification, Embeddings

BEATs – Bidirectional Encoder representation from Audio Transformers

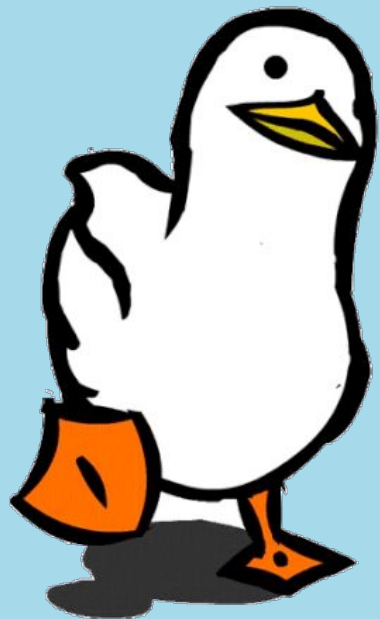
Transformer-Based Audio model that tokenizes acoustic features in recordings/data



CLAP – Contrastive Language Audio Pre-training

Maps language to audio signals. ie) Describing the taxi whistle with, “a sharp descending then ascending whistle”

Potential Challenges



- ❖ Mono-Source Data
- ❖ Research notes from competitions and other projects (using CNN models) site issues from lack of distinction between:
 - Gender
 - Audio Quality
 - Background Sounds
 - Periods of Silence



Thank You!

